

MarineGuard AI — Investigation Report

SYNTHETIC DEMONSTRATION DATA — not real-world observations.

1. Incident information

Field	Value
Incident ID	MG-2026-DEMO-001
Title	Synthetic Arabian Sea oil-slick investigation
Location	Arabian Sea, west of Mumbai (synthetic scene)
Observation time	2026-03-14T06:30:00Z
Provenance	SYNTHETIC DEMONSTRATION DATA

2. Satellite observation

SAR scene associated with this incident. Detection mode: **U-Net ML Inference**. Trained U-Net weights were used for segmentation.

3. Spill detection

Field	Value
Mode	U-Net ML Inference
Confidence	84.3%
Status	Completed

4. Spill geometry

Field	Value
Area	1650.51 km ²
Length	41.08 km
Width	40.75 km
Perimeter	163.77 km
Centroid	19.09966, 71.81835
Shape	compact

5. Probable origin

Field	Value
Latitude	19.11601
Longitude	71.74875
Estimated origin time	2026-03-14T02:00:00Z
Window	2026-03-14T00:30:00Z → 2026-03-14T03:30:00Z
Uncertainty radius	5.44 km

Probable origin is an analytical backward integration combining surface current vectors and 3% wind leeway with Coriolis deflection. It defines a probable search window, not an exact spill location.

6. Hindcast result

Backward trajectory points: 25. Model: backward_euler_drift_integration.

7. Forecast

Lead time	Latitude	Longitude
+0 h	19.09966	71.81835
+6 h	19.09163	71.92019
+12 h	19.08993	72.01336
+24 h	19.08652	72.19971
+48 h	19.0797	72.57241

Forward trajectory forecast incorporates predicted hydrodynamic surface current and 3% wind leeway. Uncertainty corridor expands over time due to oceanic turbulence and atmospheric variation.

8. AIS traffic analysis

Field	Value
Total vessels	7
Spatial candidates	7
Temporal candidates	7
Final candidates	7

9. Candidate vessel ranking

Rank	MMSI	Score	Priority
1	419000111	85.1	HIGH
2	419000222	76.8	HIGH
3	419000444	74.8	MEDIUM
4	419000555	67.3	MEDIUM
5	419000666	66.1	MEDIUM
6	419000333	65.8	MEDIUM
7	419000777	65.0	MEDIUM

10. Evidence

MT SYNTHETIC STAR (419000111) — Potential Suspect Vessel (Investigation Priority 1)

- Closest approach: 5.4 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 44.8° relative to inferred drift axis.
- Operating speed: min 0.2 kn, mean 5.0 kn. (Observed loitering/speed reduction)
- Vessel type relevance: tanker (operational context score 100%).

MV COASTAL PEARL (419000222) — Evidence-Supported Candidate #2

- Closest approach: 4.0 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 5.3° relative to inferred drift axis.
- Operating speed: min 11.8 kn, mean 12.0 kn.

- Vessel type relevance: cargo (operational context score 55%).

MV NORTHBOUND (419000444) — Evidence-Supported Candidate #3

- Closest approach: 4.6 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 28.6° relative to inferred drift axis.
- Operating speed: min 14.4 kn, mean 14.5 kn.
- Vessel type relevance: cargo (operational context score 55%).

MT FAR HORIZON (419000555) — Evidence-Supported Candidate #4

- Closest approach: 22.6 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 22.2° relative to inferred drift axis.
- Operating speed: min 16.0 kn, mean 16.1 kn.
- Vessel type relevance: tanker (operational context score 100%).

MT GAP SIGNAL (419000666) — Evidence-Supported Candidate #5

- Closest approach: 5.2 km from probable origin region.
- AIS transmissions overlap the estimated spill origin time window.
- Course consistency: Heading delta 96.5° relative to inferred drift axis.
- Operating speed: min 2.4 kn, mean 5.8 kn.
- Vessel type relevance: tanker (operational context score 100%).

11. Confidence

Detection confidence 84.3%. Drift uncertainty radius 5.44 km. Forecast corridor width 23.6 km.

12. Data gaps

- Satellite detection may include false positives, especially in prototype/demo mode.
- AIS coverage may be gapped, delayed, or spoofed.
- Ocean wind/current fields are sparse demonstration vectors, not a circulation model.

13. Responsible-AI disclaimer

This system prioritises vessels for investigation and does not establish definitive legal responsibility. The highest-ranked vessel is a highest-priority candidate only. Results require human validation and must not be used as the sole basis for legal enforcement.